සියලුම හිමිකම් ඇවිරිණි/முழுப் பதிப்புரிமையுடையது/All Rights Reserved

දකුණු පළාත් අධ්‍යාපන දෙපාර්තමේන්තුව
தென் மாகாணக் கல்வித் திணைக்களம்
Department of Education, Southern Province

13 ශ්‍රේණිය අවසාන වාර පරීක්ෂණය - 2025 (ඔක්තෝබර්)
தரம் 13 ஆண்டிறுதிப் பரீட்சை - 2025 (ஒக்டோபர்) / Grade 13 Final Term Test - 2025 (October)

Information & Communication Technology II	20	E	II	කාලය நேரம் Time	03 Hours

නම பெயர் Name ..... විභාග අංකය சுட்டிலக்கம் Index No. .....

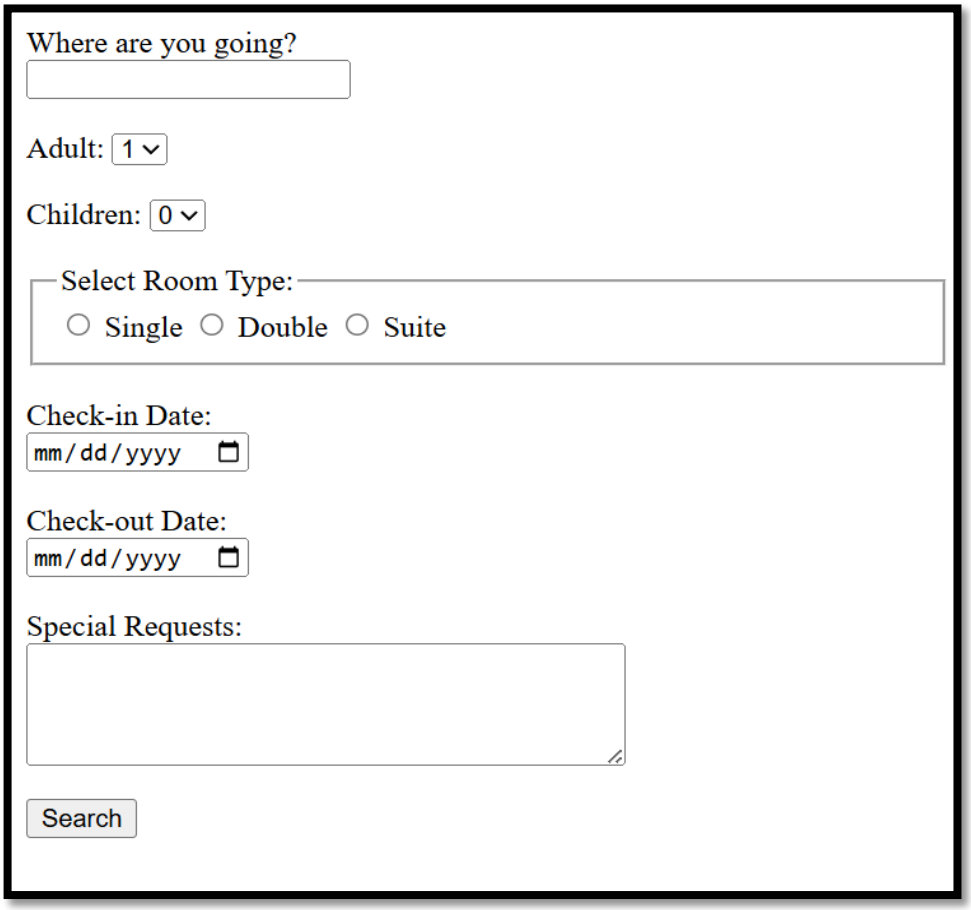
Should be considered:

- **Answer all questions in part A and only four questions selected from part B**

PART A - STRUCTURED ESSAY

Answer all four questions on this paper itself. Write your answers in the space provided for each question.

1. (a) The HTML code to get the following output is given below with some spaces.



Select the correct term from the list of terms and fill in the blanks in the code.
List of Terms: textarea, submit, fieldset, radio, /legend, method, /select, action, legend, text, select, name, /fieldset, Search, value, /textarea

Note: The same term can be used multiple times.

```
<html>
 <head></head>
 <body>
  <form _____="post" _____="action.php">
   <label for="Destination">Where are you going?</label><br>
   <input type="_____” id="Destination" name="Destination"><br><br>
   <label for="adult">Adult :</label>
   <_____ >
    <option value="1">1</option>
    <option value="2">2</option>
    <option value="3">3</option>
    <option value="4">4</option>
    <option value="5">5</option>
   <_____ ><br><br>
   <label for="children">Children:</label>
   <_____>
    <option value="0">0</option>
    <option value="1">1</option>
    <option value="2">2</option>
    <option value="3">3</option>
    <option value="4">4</option>
   <_____><br><br>

   < _____>
    < >Select Room Type:< _____>
    <input type="_____” id="single" _____="roomtype" _____="Single">
    <label for="single">Single</label>
    <input type="_____" id="double" _____="roomtype" _____="Double">
    <label for="double">Double</label>
    <input type="_____" id="suite" _____="roomtype" _____="Suite">
    <label for="suite">Suite</label>
   <_____>
  <br>

  <label for="checkin">Check-in Date:</label><br>
  <input type="date" id="checkin" name="checkin"><br><br>

  <label for="checkout">Check-out Date:</label><br>
  <input type="date" id="checkout" name="checkout"><br><br>

  <label for="requests">Special Requests:</label><br>
  <_____ id="requests" name="requests" rows="4" cols="40"></_____><br><br>

 <input type="_____” value=”_____”>
 </form></body>
</html>
```

(b) The field provided for entering the destination in this document requires its borders to be highlighted in blue to show emphasis. (attribute → border-color, value → blue)

(i) This requires adding a style using CSS, and what is the most suitable and clear style sheet type?

.....

(ii) Name the type of selector to be used to write this template.

.....

(iii) Write the pattern sheet correctly.

.....

.....

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(c) Write the output of the following code.

```
<? php
$num = 7;
$a = $num++;
$b = $num-1;
echo $b . " " . $num. " ". $a;
?>
```

.....

.....

.....

2. (a) Rewrite the following Python program using a while loop without changing the basic format of the code program to obtain the output without multiples of 5.

```
For x in range (1, 81):

        print (x)
```

.....

.....

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(b) Write down the output of the following function.

```
def display(i, j)
    L=[ ]
    while (j > 0 and i = = 0):
            x = ["*"]
            if x not in L
                    L.append (x)
            i = i - 1
            j = j + 1
    return(L)
print(dispaly(1,10))
```

.....

.....

.....

(c) Convert the following Python code into the corresponding PHP code.

```
x = int(input("Enter an Integer"))
while x < 100 :
    print(x)
    x = x + 5
print ("x is greater than 100")
```

.....

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3. (a)
In a data communication system, various encoding methods are used for data encoding. Among these encoding methods, NRZ-L (Non-Return to Zero-Level), NRZ-I (Non-Return to Zero-Inverted) and Manchester Encoding are the most special.
For the binary data block 10110010, draw the waveform encoded using the above three methods in the boxes below.

NRZ-L

NRZ-I

Manchester Encoding

(b)
Various encryption methods are used to ensure the security of information during data communication. Among them, private key encryption (Symmetric Key Encryption) and public key encryption (Asymmetric Key Encryption) occupy a prominent position.

(i) Briefly explain the difference between private key encryption and public key encryption.

……

……

……

……

(ii) What is the main challenge of private key encryption?

……

……

(iii) Provide a solution to that challenge.

……

……

(c)
Only one computer in your school's computer lab is unable to access the Internet. To solve this problem, by checking the seven layers of the OSI model step by step, the root cause of the problem can be identified accurately. Some of the tasks performed are listed below, in no particular order. Write the layers of the OSI model to which they belong.

(i) Checking whether a DNS server on the Internet can be communicated with, by using a command such as ping 8.8.8.8

……

(ii) Checking whether the computer's network cable is properly connected to the wall socket.

……

(iii) Checking whether a firewall is blocking the ports required by the web browser (e.g. 80 for HTTP, 443 for HTTPS).

……

(iv) Check if the web browser settings or proxy server settings are incorrectly configured.

……

4. (a)

(i) Write down one difference of each logical address and the physical address.

……

……

……...

(ii) Given below is a logical memory address. The main memory that accesses pages of that logical memory forms 8 frames of size 1KB.

Logical memory address

0	0	1	0	0	0	0	0	0	0	0	0	1	1

Find the number of bits available for displacement of the physical memory address.

……

……

……

(iii) In the logical memory reference,
Write down the page number: (A) ……. … Displacement: (B) ……

(iv) A part of the page table used for the above paging process is given below.

Page number	Frame number	Yes / No bit
0	1	0
1	7	1
2	1	1
3	5	1
4	4	1

According to the page table, find the physical memory address accessed by the above logical address.

……

……

……..

(b) (i) Write the reason why a process running in a central processing unit terminates the running state and enters the following process state.

Ready state: - …… Blocked state:- …….

(ii) What is the other state that a process can enter after exiting the running state?

……

Name the most common reason for that

……

PART B- ESSAY

Answer only four questions from part B.

5. (a)
For a system to operate, it must input power using three input parameters X, Y, Z. When the total input power is greater than or equal to 40 watts, the system is active (logical value 1) and the power value provided during operation for each input is given in the table below (the power input to each parameter is logical value 1).

Input	power value (watts)
X	10
Y	20
Z	30

This system operates only when an input combination is provided that satisfies the power requirement.

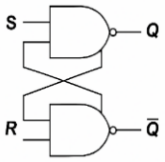
(i) Prepare a truth table for the above description.

(ii) Write down a sum of products (SOP) expression for the truth table obtained in (i) above.

(iii) Obtain the simplest expression for the SOP expression obtained in (ii) above.

(iv) Represent the simplest expression obtained in (iii) above using only NOR gates

(b) The figure below shows an SR Latch made using NAND gates.



(i) What are the output values of Q and $\overline{Q}$ of this SR Latch when S = 1, R = 0?

(ii) Write down the values of Q and $\overline{Q}$ when S = 0, R = 1.

(iii) Given the values of S = 0, R = 0, what is the output?

(iv) Briefly describe the state of the SR Latch when S = 1, R = 1.

6. (a) (i) What is Public Switched Telephone Network (PSTN)?

(ii) Fill in the blank with the suitable answer.

.....(1).....transmitting a signal into .....(2).....signal through telephone lines is called .....(3)....., and the recipient of the message again converts .....(4).....signal into .....(5).....signal is called .....(6)....., and for this purpose .....(7)..... is used. However, this device is not required to transmit data through a medium that can transmit .....(8)..... signals.

(b) (i)

- There are two local area networks (LANs) L1 and L2. The L1 network has a domain name server (DNS server), a web server, a mail server, a computer (C1) and a network printer. The L2 network has four computers (C1, C2, C3 C4) and a printer.
- Two network topologies, L1 and L2, are created using two network topologies. The L1 network is designed as a star network topology, and each computer in the L2 network is connected to two computers on either side of it.
- An Internet Service Provider (ISP) provides internet access to this entire network and a firewall is connected to the network.

Draw the network diagram of this network.

Note - Clearly name the types of switches and devices used in the network.

(ii) Write down the function of a mail server computer and two protocols used.

(c) Bag Import & Export (pvt) Ltd is assigned the IP address 173.23.68.0/24. The company intends to create three subnets, B1, B2, and B3. Each subnet has 31, 27, and 5 computers, respectively, and a network printer is installed in B1 and a printer in B3.

(i) Write the Subnet mask of the IP address assigned to the institution in Dotted Binary Notation.

(ii) Write the Network Address, Broadcast Address, First usable IP Address and Last usable IP Address for each of the above subnets.

(d) A simple method of checking whether the data being transmitted has been changed during data transmission is called Parity Check.

(i) If the data 111010101 is transmitted using the parity check method, write down the parity check method and explain how it is done.

(ii) Is there a problem with the method you mentioned above? Explain it.

7. (a) (i)

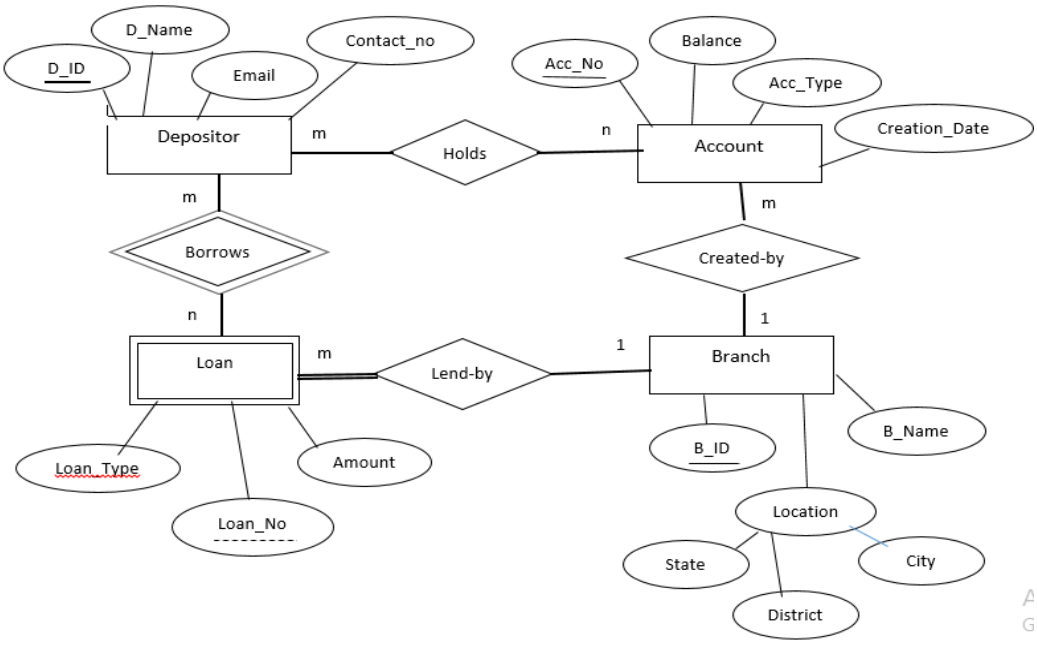
The student who receives the subject prize (Subject_Prize) for each subject in each grade is selected based on the highest average of the marks obtained by the students in the three term tests of the classes corresponding to grades 6–13 in a school year. A student may be eligible for several subject prizes. Several students may receive one subject prize. A prize number (Prize_ID) is used to uniquely identify a prize. A class name (Class_Name) is used to uniquely identify a class.

The student who obtains the highest average of the total marks obtained in the three term tests among the students of that class is eligible for the class prize (Class_Prize) in the classes related to each grade. A student in a class can receive only one class prize and one class prize can be awarded to several students in that class. There can be several class prizes in a grade and one class prize belongs to only one grade.

An admission number (Index_No) is used to uniquely identify a student and data such as the student's name (Name) as initials (Initial) and surname (Surname), date of birth (DOB), address (Address), telephone numbers (Tel_No) are stored. There is a subject number (Sub_ID) associated with a subject, which can be used to uniquely identify the subject. Information about the subject name (Sub_Name) and the teachers teaching the subject (Sub_Teacher) is stored. Each grade is uniquely identified by its number (Grade_ID) and there is a grade head (Grade_Head) for each grade.

Draw an ER Diagram for the above description. If you have any assumptions, clearly state them.

(ii) Consider the following ER Diagram related to a banking system. Write down its Relational Schema.



(b) Consider the following relation in a database belonging to an institution that conducts IT courses.

Student – Course

Student_ID	Course_ID	Course_Name	Teacher_Name	Teacher_Tel
S0001	IT1100	Networking	Nanayakkara	0712123100
S0004	IT1101	Web Designing	Alahakoon	0777700250
S0005	IT1102	Information System	De Silva	0747526374
S0003	IT1103	Cyber Security	Nimali	0715923321
S0009	IT1104	Database	Jayawardana	0763924257
S0010	IT1103	Cyber Security	Nimali	0715923321
S0002	IT1100	Networking	Nanayakkara	0712123100

(i) In which normal form is the above Student-Course table? Give reasons for your answer.

(ii) Convert the above table into third normal form (3NF) tables.

(c) Consider the following table from a database maintained in a school library.

Book_Details

Book_Code	Title	Author's_Name	Number_of_Books
2131	Internet	Mahesh Gankanda	10
2132	Computer Studies	Suranimala Basnayaka	5
2133	Essay Writing	Nimali Bogoda	12
2134	Education Psycology	D. Jayasinghe	6
2135	Tamil Language	Raj Sudakaran	8
2136	Computer Studies	Mohomad Niyas	4
2137	English Tamil Dictionary	Raj Sudakaran	6

(i) After creating the above Book_Details table, write down the SQL code to insert the field "Date of Published" into that table.

(ii) Write down the SQL code to insert the data into the above Book_Details table, such that the Book_Code value is 2141, the Title value is Education Technology, the Author's_Name value is D.P. Athukorala, and the Number of Books is 9.

8.

To create a system for a park for entertaining games, the information obtained by the system analysts in discussion with the service candidate is given below.

Every person who comes to the park must register by providing personal information at the registration counter before playing water games, and that information is stored in a computerized personal information table. After personal registration, a unique QR code is provided to each registered person by the registration counter.

To enter the park and play water games, the QR code in his/her possession must be provided to the relevant game counter. Information about the water games performed by each registered person is stored in a temporary game information database.

In calculating the price, the charge for each game is obtained from the computerized game price table. When the game is finished, when the person goes to the payment counter and gives his/her QR code, the final bill is calculated and given considering the water games performed during the time he/she was in the park. Once the relevant payment is made, a receipt will be issued confirming that the relevant payment has been made.

(a) Create a Data Flow Diagram (DFD) using the above information.

(b) Write down two functional requirements and two non-functional requirements for this system.

9. (a) Draw a flowchart to allow a user to input 10 positive integers of his choice, output the multiples of two of those 10 integers separately, and finally output the average of the multiples of the two.

(b) In the above algorithm, if the user is expected to provide -1 as an input to exit the program at any time, write the suitable Python program for that purpose, including that change.

(c) Build a Python function corresponding to the bubble sort method for arranging a sequence of numbers in descending order.

10. (a)

The establishment of a smart traffic management system in a city is expected to reduce traffic congestion, improve travel times, and improve overall traffic efficiency.

This smart traffic management system uses multiple AI agents to optimize the traffic flow in the city.

The smart traffic management system includes traffic light agents, navigation agents, and emergency vehicle agents.

The traffic light agent's job is to adjust traffic light signal timing based on real-time traffic data received from sensors and other agents. The navigation agent provides real-time traffic updates and suggests alternative routes to the driver through the navigation app. The emergency vehicle agent prioritizes emergency traffic signals, ensuring fast response times.

Draw a simple agent diagram for the multi-agent system shown above.

(b) (i) Briefly explain the concept of Ubiquitous Computing.

(ii) Name three technologies used for Ubiquitous Computing.

(iii) Name two applications of Ubiquitous Computing and write a special feature of each application.

(c) (i) Write two advantages of e-commerce.

(ii) Name four applications of e-commerce.

(iii) Name two special features of mobile commerce.

(iv) Write two disadvantages of mobile commerce.